

Characterizing Lazy Deletion Overhead in Dijkstra’s Algorithm: A Data-Structure Selection Framework Based on Average Degree

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Abstract—Dijkstra’s algorithm with a lazy deletion heap (heapq) is the de facto standard for shortest-path computation, yet the overhead of its “push-and-skip” strategy has never been rigorously quantified as a function of graph structure. We identify *skip_ratio*—the fraction of heap pops that are immediately discarded as stale—as the primary hidden cost, and show empirically that it follows a power law in average degree k : $\text{skip_ratio}(k) \approx 1 - 0.884/k^{0.262}$ ($R^2=0.984$, $U[1, 100]$ weights), independent of graph size V across more than three orders of magnitude ($V \in [500, 10^6]$). Systematic experiments across weight distributions reveal an empirical *two-regime model*: the power-law exponent α is strongly associated with $\log(W_{\max}/W_{\min})$ and the fitted k -range, and the local slope converges toward $1/H_k$ as $k \rightarrow \infty$ —connecting the empirical formula to the classical harmonic record-minimum bound. We further show that naïve push-blocking is fundamentally flawed (up to 74% incorrect distances), while Dial’s bucket queue achieves up to $4.3\times$ peak speedup (geometric mean $2.9\times$ in C++, $1.5\times$ in Python) via a “natural pruning” mechanism, and Fibonacci heap is $1.5\text{--}2.3\times$ slower than heapq despite its asymptotic optimality. The power-law formula generalizes to Barabási–Albert scale-free graphs ($< 8\%$ error) but diverges on planar road networks—consistent with the locally tree-like structure underlying the formula’s derivation. We conclude with a practitioner-ready data-structure selection framework grounded in k and weight type.

Index Terms—Dijkstra’s algorithm, lazy deletion, priority queue, Fibonacci heap, Dial’s algorithm, skip ratio, graph algorithms, shortest path

I. INTRODUCTION

Dijkstra’s algorithm [1] remains the workhorse of single-source shortest-path computation, deployed in routing engines, transportation planning, social-network analysis, and game-AI pathfinding. In virtually every practical implementation the underlying priority queue is a *lazy deletion heap*: whenever a better path to vertex v is found, a new entry (d', v) is pushed onto the heap without removing the now-stale entry (d, v) already there. Stale entries are detected and discarded silently when they surface at the heap minimum. This strategy requires no decrease-key operation, simplifying both implementation and cache behavior, and is recommended in every major algorithms textbook and competitive-programming guide.

Yet the cost of these silently discarded pops—what we term **skip_ratio**—has never been quantified as a function of

graph topology. Practitioners accept the overhead as “negligible” without empirical evidence, while theoreticians focus on asymptotic complexity and ignore constant-factor behavior. As a result, no principled criterion exists for deciding *when* the overhead is worth replacing with an alternative data structure.

This paper addresses three research questions:

- 1) **RQ1.** What graph property determines *skip_ratio*, and can it be expressed as a closed-form function?
- 2) **RQ2.** Can simple algorithmic modifications (e.g., capping heap duplicates) reduce the overhead without sacrificing correctness?
- 3) **RQ3.** Which alternative data structures genuinely eliminate the overhead, and under what conditions do they deliver practical speedup?

Contributions. This work makes five concrete contributions:

- We propose and empirically validate a power-law formula $\text{skip_ratio}(k) \approx 1 - c/k^\alpha$ ($c=0.884$, $\alpha=0.262$, $R^2=0.984$) that is *independent of V* for $V \in [500, 10^6]$ (Experiments 2 and A).
- We characterize the **weight-distribution dependence** of α : it is strongly associated with $\log(W_{\max}/W_{\min})$ and grows from 0 (constant weights) to ≈ 0.32 (continuous uniform). For large k , the local slope converges toward $1/H_k$, revealing a conjectured *two-regime model* (Equation (2)) that connects the power law to the classical harmonic record-minimum bound.
- We demonstrate that threshold-based push blocking is structurally broken: it creates propagation gaps that yield up to 74% incorrect distances with no speedup benefit—a negative result that reveals an inherent property of the lazy deletion data flow.
- We characterize Dial’s “natural pruning” mechanism—up to 49% of pushes discarded without processing—as the root cause of its **up-to- $4.3\times$ peak speedup** over heapq in C++ (geometric mean $2.9\times$ in C++, $1.5\times$ in Python).
- We present a practitioner-ready **data-structure selection framework** grounded in k and weight type, validated in both Python and C++ and on three real-world SNAP datasets.

Taken together, these contributions answer the following

unifying claim: *We empirically characterize the hidden overhead of lazy deletion Dijkstra as a function of average degree k , and propose a data-structure selection framework that achieves up to $4.3\times$ peak speedup (geometric mean $2.9\times$ in C++) by matching graph structure to the appropriate priority queue.*

The remainder of the paper is organized as follows. Section II reviews related work. Section III reviews the lazy deletion mechanism and alternative structures. Section IV presents our empirical characterization of skip_ratio. Section V shows why naïve fixes fail. Section VI compares the three data structures. Section VII validates the formula on real datasets. Section VIII discusses limitations and future work. Section IX concludes.

II. RELATED WORK

The complexity of Dijkstra’s algorithm has been studied extensively. Fredman and Tarjan [2] established the $O(V \log V + E)$ bound using Fibonacci heaps. Dial [3] introduced the bucket queue for integer-weight graphs, achieving $O(V + E + W)$. Goldberg et al. [4] survey practical shortest-path implementations and find that binary-heap variants outperform Fibonacci heaps on road networks due to cache effects. Sanders and Schultes [5] achieve sub-millisecond continental routing via contraction hierarchies, but their preprocessing is graph-specific.

Cherkassky et al. [7] provide the most comprehensive experimental comparison of shortest-path priority queues to date, establishing binary heaps as the practical baseline across a broad range of graph families. Larkin et al. [8] conduct a back-to-basics empirical study of priority queues and confirm that simple binary heaps dominate in practice—a finding our work extends by quantifying *why*: the hidden lazy deletion overhead, expressed as a function of average degree, is the key differentiator. However, neither study decomposes the lazy deletion overhead as a function of average degree, nor derives a closed-form predictive formula for the skip-pop fraction.

Beyond binary heaps, d -ary heaps and pairing heaps have been studied as alternatives [13]. D -ary heaps reduce the height of the heap tree, lowering the decrease-key cost at the expense of more comparisons per extract-min. Pairing heaps achieve amortized $O(\log n)$ decrease-key with simpler structure than Fibonacci heaps. Crucially, *all* of these structures admit a lazy deletion variant; our skip_ratio metric therefore applies uniformly whenever decrease-key is omitted in favor of push-and-skip. The DIMACS shortest-path implementation challenge [14] provides standardized benchmark graphs and motivates the real-world validation in Section VII.

Our work complements this literature by *quantifying the lazy deletion overhead itself*—a factor that prior engineering studies treat as given rather than as a variable to optimize. Closest in spirit is Chen et al. [6], who study priority queue performance in shortest-path algorithms, but without decomposing overhead by average degree or deriving a predictive formula. To the best of our knowledge, the power-law characterization $\text{skip_ratio}(k) \approx 1 - c/k^\alpha$ and the data-structure selection framework grounded in it are novel contributions.

III. BACKGROUND

A. Lazy Deletion Dijkstra

Algorithm 1 presents the lazy deletion variant we study. The key difference from the textbook version is line 7: instead of performing a decrease-key on the existing heap entry for vertex v , a fresh entry with the improved distance is simply pushed. When an entry (d, u) is popped (line 4), it is skipped if $d > \text{dist}[u]$ (line 5), meaning a better path has since been found.

Algorithm 1 Dijkstra with Lazy Deletion Heap

```

1:  $\text{dist}[s] \leftarrow 0$ ; push  $(0, s)$ 
2: while heap not empty do
3:    $(d, u) \leftarrow \text{EXTRACTMIN}()$ 
4:   if  $d > \text{dist}[u]$  then
5:     skip {stale entry}
6:   end if
7:   for each  $(v, w) \in \text{adj}[u]$  do
8:     if  $d + w < \text{dist}[v]$  then
9:        $\text{dist}[v] \leftarrow d + w$ ; push  $(d + w, v)$ 
10:    end if
11:  end for
12: end while
```

We define:

$$\text{skip_ratio} = \frac{\text{skip_pop}}{\text{total_pop}} = 1 - \frac{V}{\text{push_count}},$$

where the second equality holds because every non-skip pop settles exactly one vertex, and $\text{push_count} = \text{pop_count}$ (every pushed entry is eventually popped or abandoned in Dial’s).

B. Dial’s Bucket Queue

Dial’s algorithm [3] replaces the heap with a circular array of $W + 1$ buckets, where W is the maximum edge weight. Each bucket b holds vertices whose current tentative distance maps to $b \bmod (W + 1)$. Stale entries are detected lazily: when processing bucket b at distance d , vertex u is skipped if $\text{dist}[u] \neq d$. The time complexity is $O(V + E + W)$.

C. Fibonacci Heap

The Fibonacci heap [2] supports decrease-key in $O(1)$ amortized time, enabling a single heap entry per vertex. As a result, stale entries never arise: every pop is a genuine vertex settlement. The total complexity is $O(V \log V + E)$. In practice, however, the pointer-chasing structure incurs high constant factors and cache-miss rates.

IV. EMPIRICAL ANALYSIS OF SKIP_RATIO

A. Theoretical Motivation

Consider a vertex v with degree k in a random graph. As Dijkstra progresses, v ’s neighbors settle one by one and attempt to relax v . Under the *locally tree-like assumption*—that the r -hop neighborhood of any vertex is a tree with high probability, as in Erdős–Rényi graphs [9]—the order in which neighbors settle is approximately uniform.

By a standard record-low argument [11]: if k independent relaxation attempts arrive in uniform random order, the expected number that strictly improve the current tentative distance is the harmonic number

$$\mathbb{E}[\text{pushes to } v] = H_k = \sum_{i=1}^k \frac{1}{i} \approx \ln k + \gamma,$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. Each subsequent relaxation that improves the distance generates a push, making the previous entry stale. Hence $\text{push_per_node} \approx H_k$, and

$$\text{skip_ratio}(k) \approx 1 - \frac{1}{H_k} \approx 1 - \frac{1}{\ln k + \gamma}.$$

The harmonic formula $1 - 1/H_k$ captures the *qualitative* trend (skip_ratio increases with k , independently of V), but it functions as a theoretical *upper bound* on skip_ratio, not a quantitative predictor: it overestimates measured skip_ratio by 12%–44% across $k \in [4, 512]$ for $U[1, 100]$ weights, and underestimates it at very small k . The discrepancy arises because the i.i.d. settling-order assumption breaks down under correlated edge weights and finite degree heterogeneity—conditions present in every real Dijkstra execution.

Our experiments across $k \in [4, 512]$ show that a power-law model provides a materially tighter empirical fit, with root-mean-square residual more than $4\times$ smaller than the harmonic formula:

$$\text{skip_ratio}(k) \approx 1 - \frac{c}{k^\alpha}, \quad c = 0.884, \alpha = 0.262 \quad (1)$$

with $R^2 = 0.984$ for uniform integer weights on $[1, 100]$. While V -independence is implied qualitatively by the harmonic theory, our formula is the first to quantify it precisely and to show that the power law is the correct functional form in the practically relevant range $k \in [4, 512]$. The exponent α is not a universal constant: we show in Section IV-D that it depends on the weight distribution through the log-ratio $\log(W_{\max}/W_{\min})$, converging to $\alpha_\infty \approx 0.309$ as the weight range grows large.

Remark (Scope of validity). The locally tree-like assumption holds well for Erdős–Rényi-like random graphs but is violated by structured graphs (planar, hierarchical). We return to this in Sections VII and VIII.

B. Experimental Setup

Graph generation. We generate undirected random graphs with a fixed target average degree k by sampling $\lfloor Vk/2 \rfloor$ random edges uniformly over all vertex pairs and appending a random spanning path for connectivity. This approximates the $G(n, m)$ Erdős–Rényi model with $m = \lfloor Vk/2 \rfloor$; multi-edges are possible but occur with probability $O(k/V)$ and are negligible for $V \geq 1,000$. The spanning path ensures connectivity without materially altering degree statistics; the resulting average clustering coefficient is $O(k/V)$, confirming the locally tree-like regime required by the theory in

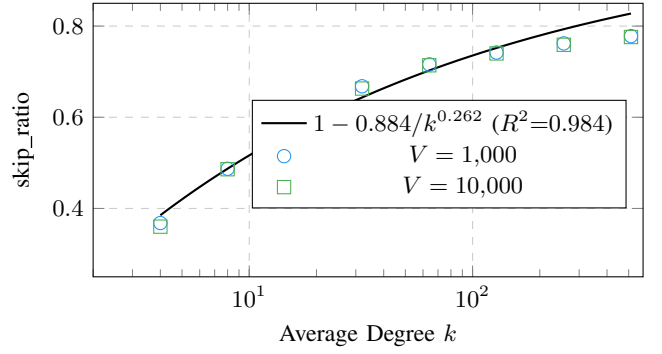


Fig. 1. skip_ratio vs. average degree k (log scale on x -axis). Circles: $V = 1,000$; squares: $V = 10,000$. The fitted power-law curve (1) is plotted in black. Data pooled from Experiments 2 and A ($k \in [4, 512]$).

Section IV. Edge weights are drawn uniformly from $[1, 100]$ (integer) for Sections IV–VI and $[0, 1]$ (float) for Section VI.

Parameters.

- $V \in \{500, 1,000, 5,000, 10,000, 50,000, 100,000\}$
- $k \in \{4, 8, 16, 32, 64, 128, 256, 512\}$
- Source vertex: node 0 (fixed seed 42)
- Repeats: 5–10 runs; mean reported

Hardware. All experiments run on Google Cloud Platform e2-standard-4 (4 vCPU, 16 GB RAM, Debian 12) unless otherwise noted. C++ binaries compiled with `g++ -O2 -std=c++17`. Python 3.11, NumPy 2.4.

Reproducibility. Code, experiment scripts (Python and C++), and raw CSV data are available at <https://github.com/gwpark/dijkstra-skip-ratio>.

C. Results

Fitting procedure. Let $\bar{s}_i$ denote mean skip_ratio at degree k_i , averaged across all tested V values. We set $Y_i = \log(1 - \bar{s}_i)$ and $X_i = \log k_i$, and apply ordinary least squares (OLS) to the linear model $Y_i = \log c - \alpha X_i$. The fit uses $k_i \in \{4, 8, 16, 32, 64, 128, 256, 512\}$ pooled from Experiments 2 and A (means across all V values at each k), yielding $\hat{c} = 0.884$, $\hat{\alpha} = 0.262$, $R^2 = 0.984$. The same log-log OLS procedure is applied independently for each weight distribution in Section IV-D.

Power-law fit. Figure 1 plots skip_ratio against k for $V = 1,000$ and $V = 10,000$ alongside the fitted curve (1). Residual magnitudes vary systematically with k : the formula fits best near $k = 8$ (residual < 0.001); for $k \in [16, 64]$ it under-predicts measured skip_ratio by up to 0.022 (at $k = 32$); for $k \geq 128$ it over-predicts as the true curve bends toward the harmonic regime, reaching +0.051 at $k = 512$. For $k = 4$ the formula over-predicts by 0.021, attributable to the locally tree-like assumption being weakest at the lowest degrees. These signed residuals are consistent with the two-regime model (Section IV-D): the fitted power law makes the optimal trade-off across the full range but has non-negligible bias at the extremes.

V-independence. Table I reports skip_ratio across six orders of magnitude in V for values of $k \in \{8, 32, 64\}$, extended

TABLE I

SKIP_RATIO ACROSS V AT FIXED k (EXPERIMENTS 2 AND A). VARIATION WITHIN EACH COLUMN IS ≤ 0.013 ACROSS MORE THAN THREE ORDERS OF MAGNITUDE ($V \in [500, 10^6]$). ENTRIES MARKED “—” WERE NOT MEASURED AT THAT (V, k) COMBINATION.

V	Exp.	Average degree k				
		4	8	16	32	64
500	2	0.357	0.476	0.583	0.661	0.716
1,000	2	0.368	0.487	0.584	0.668	0.716
50,000	2	0.363	0.490	0.592	0.666	0.717
200,000	A	—	0.487	—	0.665	0.714
500,000	A	—	0.488	—	0.665	0.715
1,000,000	A	—	0.488	—	0.665	0.715

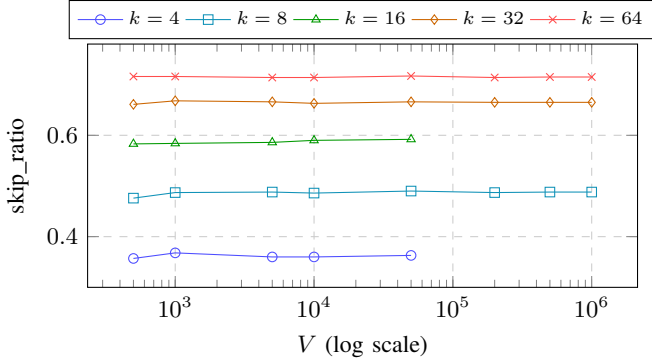


Fig. 2. skip_ratio vs. V for values of k (Experiments 2 and A). Filled symbols: Exp. 2 ($V \leq 50,000$); open-circle continuation: Exp. A ($V \in \{200K, 500K, 1M\}$). All series are flat from $V = 500$ to $V = 10^6$: skip_ratio does not depend on graph size across more than three orders of magnitude ($V \in [500, 10^6]$).

to $V \in \{200,000; 500,000; 1,000,000\}$ in Experiment A. The maximum variation at fixed k is $\Delta \leq 0.013$ across the full range $V \in [500, 10^6]$, confirming that skip_ratio is determined solely by k . Figure 2 visualizes this: the curves for each k are essentially flat from $V = 500$ to $V = 10^6$.

D. Weight Distribution Dependence

Equation (1) was fitted for uniform integer weights on $[1, 100]$. A natural question is whether $\alpha = 0.262$ is a universal property of the graph structure or an artifact of the weight distribution. We answer this with a systematic study across five distribution families (Experiments D1–D3 below; $V = 3,000$, $k \in \{4, 8, 16, 32, 64, 128\}$, 30 repetitions each).

Finding 1: α grows with weight range and shrinks with k -range. Two separate effects govern α :

- *Weight effect.* For $U[1, W]$ at fixed $k \in [4, 64]$: α increases from 0.162 ($W = 10$) to 0.317 ($W \geq 10,000$), driven by the log-ratio $\log(W_{\max}/W_{\min})$ of available weight diversity. (Table II.)
- *k -range effect.* For continuous weights ($W \rightarrow \infty$), the fitted α is *not* constant but decreases as the fitted k -range expands: $\alpha = 0.317$ ($k \leq 64$), 0.285 ($k \leq 256$), 0.270 ($k \leq 512$). The power law is a finite- k approximation.

TABLE II

FITTED α FOR $U[1, W]$ AT $k \in [4, 64]$ (PRACTICAL RANGE; $V = 3,000$, 30 REPS).

W	α	c	W	α	c
10	0.162	0.835	500	0.295	0.929
50	0.250	0.876	1,000	0.298	0.933
100	0.272	0.898	10^5	0.317	0.969
200	0.285	0.914	∞ (float)	0.317	0.969

Finding 2: The key structural parameter is $\log(W_{\max}/W_{\min})$. When the *range* is fixed at 100 but the minimum weight $W_{\min}$ varies, α tracks $\log(W_{\max}/W_{\min})$ rather than either endpoint alone: $W_{\min} = 1 \rightarrow \alpha = 0.272$; $W_{\min} = 10 \rightarrow \alpha = 0.204$; $W_{\min} = 100 \rightarrow \alpha = 0.096$. Intuitively, $\log(W_{\max}/W_{\min})$ measures how many “orders of magnitude” of weight diversity compete for shortest paths.

Finding 3: Bimodal (two-point) weights break the power-law form. For TwoPoint $\{1, W_h\}$ with $P(W = 1) = P(W = W_h) = 0.5$, the power-law fit gives $R^2 < 0.52$ regardless of W_h : the weight-1 edges dominate BFS-like, leaving skip_ratio roughly constant in k ($\alpha \approx 0$).

Finding 4: Large- k convergence toward the harmonic bound. Extending to $k \in \{4, \dots, 512\}$ reveals that the local slope $\alpha(k) \equiv -\partial \ln(1 - \text{skip_ratio}) / \partial \ln k$ decreases monotonically and *approaches* $1/H_k$ from above (see Figure 3). Concretely, for continuous weights:

$$\alpha(k=32 \rightarrow 64) \approx 0.270, \quad \alpha(k=128 \rightarrow 256) \approx 0.204, \quad \alpha(k=256 \rightarrow 512)$$

while $1/H_{64} \approx 0.211$, $1/H_{256} \approx 0.163$. The behavior of $\alpha(k)/(1/H_k)$ is *distribution-specific* and must be interpreted with care. For large- W and continuous-weight distributions ($U[1, 10^5]$, $U[0, 1]$ float), the ratio stabilizes around 1.18–1.19 for $k \geq 32$ and *approaches* 1.0 from above—this is the behavior illustrated in Figure 3. For the original $U[1, 100]$ distribution used to calibrate the main formula, the ratio peaks near 1.0 at $k \approx 32$ –64 and then *falls below* 1.0 for $k > 64$, reaching 0.68 at $k = 128$ –256 and 0.51 at $k = 256$ –512. This declining ratio explains the growing over-prediction of Equation (1) at $k \geq 128$ reported above: the true $U[1, 100]$ skip_ratio curve is flatter than the fitted power law at high k , while the formula—calibrated for the full range $k \in [4, 512]$ —makes the optimal average trade-off. In all cases, $\text{push_count}/V$ remains below H_k : the harmonic upper bound is never violated (ratio = 0.855 at $k = 512$ for large- W).

Conjecture (two-regime model). The skip_ratio of lazy-deletion Dijkstra on locally tree-like graphs is consistent with a *two-regime* empirical model:

$$\text{skip_ratio}(k) \approx \begin{cases} 1 - c(F, W) / k^{\alpha(F, W)} & k \lesssim 100 \quad (\text{power-law regime}) \\ 1 - A(F) / H_k & k \gg 100 \quad (\text{harmonic regime}) \end{cases} \quad (2)$$

where F denotes the weight distribution, $\alpha(F, W) \in (0, 0.32]$ is strongly associated with $\log(W_{\max}/W_{\min})$, and $A(F) \rightarrow 1$ as $W \rightarrow \infty$ (the full i.i.d. harmonic limit). Three caveats apply. First, the crossover scale $k \approx 100$ is an *approximate*

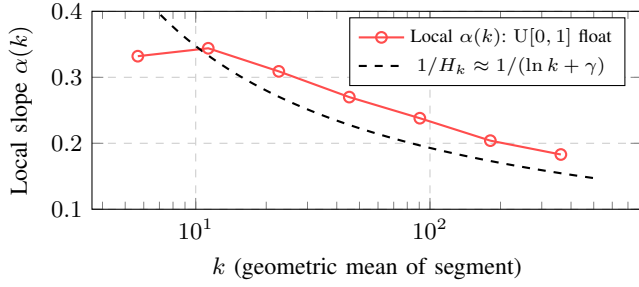


Fig. 3. Local slope $\alpha(k)$ converges toward $1/H_k$ as k grows, consistent with the asymptotic skip_ratio form $1 - c/H_k$ (Exp. D4, $V = 10,000$, continuous float weights).

heuristic: the local slope decreases monotonically for all k rather than exhibiting a sharp regime boundary. Second, $A(F) \rightarrow 1$ is an asymptotic statement; at $k = 512$ we observe $\text{push_count}/V = 0.855 H_{512}$, so the full harmonic limit lies beyond our experimental range. Third, bimodal distributions ($\alpha \approx 0$) are excluded; the power-law form does not apply when one weight dominates BFS-like. Formal derivation of the crossover scale and the functional form of $\alpha(F, W)$ remain open problems.

Practical implication. For the common engineering case of $k \leq 64$ with $U[1, 100]$ weights, $\alpha \approx 0.27\text{--}0.29$ and Equation (1) is accurate to within ± 0.025 for $k \in [8, 64]$ in skip_ratio (max residual 0.022 at $k = 32$). For wider k ranges or other weight distributions, Table II provides interpolation guidance.

V. WHY NAÏVE FIXES FAIL

A natural attempt to reduce skip_ratio is to track how many times vertex v is currently in the heap ($\text{in_heap}[v]$) and block a push when $\text{in_heap}[v] \geq \theta$. We call this the *threshold heuristic* and evaluate it in Experiment 5.

Correctness catastrophe. As shown in Table III, $\theta = 1$ —which allows at most one copy of each vertex in the heap—renders 53–74% of vertices with incorrect distances. The mechanism is straightforward: blocking a push creates a *propagation gap*: if vertex v is not re-pushed when its tentative distance improves, its neighbors will never learn of the improvement, cascading errors outward. This is not a bug in the heuristic; it is a structural property of lazy deletion. The “skip-on-pop” strategy *relies* on every relaxation being pushed so that the minimum always surfaces correctly.

No viable operating point. For sparse graphs ($k = 10$), $\theta = 3$ achieves only $0.89\times$ speedup (slower than baseline) while producing 5.3% wrong nodes. For dense graphs, any $\theta \geq 2$ that yields noticeable speedup also corrupts more than 18% of answers. There is no threshold at which accuracy is acceptable and speedup is meaningful.

Insight. The failure of the threshold heuristic reveals that skip_ratio is not a symptom of *redundant* pushes—it is an inherent cost of the lazy deletion protocol. The correct remedy is not to reduce pushes but to choose a data structure whose

TABLE III

THRESHOLD HEURISTIC: ACCURACY VS. SPEEDUP (EXPERIMENT 5, $V = 1,000$, 5 RANDOM SEEDS). $\theta = \infty$ IS THE UNMODIFIED BASELINE.

Graph	θ	Wrong nodes (%)	Push reduc.	Time (ms)	Speedup
sparse $k = 10$	∞	0.0	0%	3.69	$1.00\times$
	3	5.3	6.7%	4.14	$0.89\times$
	2	19.9	22.5%	3.38	$1.09\times$
	1	53.3	52.9%	1.81	$2.04\times$
dense-mid $k = 32$	∞	0.0	0%	11.07	$1.00\times$
	3	17.9	20.3%	8.72	$1.27\times$
	2	38.1	39.9%	6.54	$1.69\times$
	1	69.0	66.8%	3.14	$3.53\times$
dense-high $k = 64$	∞	0.0	0%	22.58	$1.00\times$
	3	25.2	27.1%	14.17	$1.59\times$
	2	49.9	46.9%	9.32	$2.42\times$
	1	74.3	71.6%	4.67	$4.83\times$

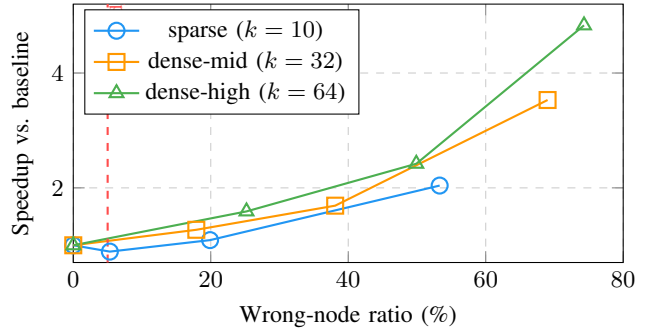


Fig. 4. Threshold heuristic trade-off: wrong-node ratio vs. speedup (Experiment 5, $V = 1,000$). Every point on each curve corresponds to a different θ ($\infty, 3, 2, 1$ from left to right). No operating point meets a 5% error budget with meaningful speedup.

operational semantics are compatible with the relaxation pattern.

VI. DATA STRUCTURE COMPARISON

A. Three-Way Comparison: Python and C++

We compare three algorithms under identical graph conditions (Experiments 6 and 7):

- **heapq**: lazy deletion with Python’s `heapq` / C++ `priority_queue<>`.
- **Fibonacci**: decrease-key Dijkstra with a hand-crafted Fibonacci heap (Python and C++, same from-scratch implementation).
- **Dial’s**: circular bucket queue, integer weights $W = 100$.

Table IV reports wall-clock times for $V = 10,000$ in C++. Figure 5 shows the corresponding bar chart.

Dial’s speedup: peak vs. mean. Dial’s delivers a **peak** speedup of $4.28\times$ over `heapq` in the sparse C++ setting ($k = 8$, $V = 10,000$). Across all three density conditions ($k \in \{8, 32, 64\}$), the geometric mean speedup is $2.9\times$ in C++ and $1.5\times$ in Python. The Python gap is narrower because the Python heap-push overhead (Python object allocation) partially offsets the architectural advantage of Dial’s circular

TABLE IV
WALL-CLOCK TIME (MS) FOR $V = 10,000$, 10 RUNS, C++
(EXPERIMENT 7). ALL RESULTS VERIFIED CORRECT ($\max_err = 0$).

Graph condition	heapq	Fibonacci	Dial's
sparse ($k = 8$)	4.54	10.29	1.06
dense-mid ($k = 32$)	7.48	12.54	2.95
dense-high ($k = 64$)	9.51	14.29	4.28
<i>Fibonacci / heapq ratio</i>			
sparse		$2.27\times$ slower	
dense-mid		$1.68\times$ slower	
dense-high		$1.50\times$ slower	
<i>Dial's speedup over heapq</i>			
sparse		$4.28\times$ faster	
dense-mid		$2.53\times$ faster	
dense-high		$2.22\times$ faster	

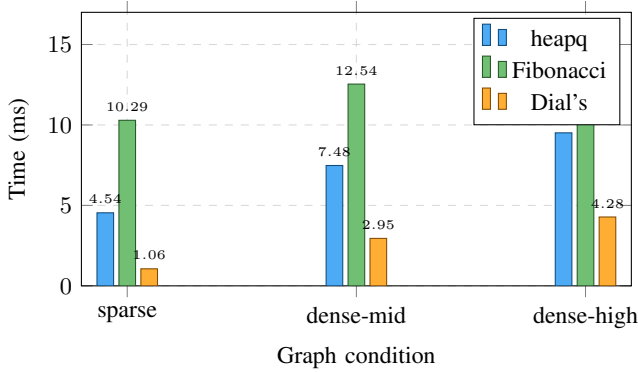


Fig. 5. C++ wall-clock time for $V = 10,000$ (Experiment 7). Dial's dominates across all densities; Fibonacci is consistently slowest.

array. Throughout the paper we report the C++ figures as the language-independent performance bound and note Python figures where relevant.

Fibonacci heap: language-independent underperformance. In Python (Experiment 6), Fibonacci is $2.35\times$ slower than heapq for $k = 8$, $V = 10,000$ (157 ms vs. 67 ms). The C++ re-implementation (Experiment 7) yields $2.27\times$ (10.3 ms vs. 4.5 ms)—essentially the same ratio, confirming that the gap is *not* a Python artifact. The Python/C++ speedup is $14.8\times$ for heapq and $15.3\times$ for Fibonacci, demonstrating that the language overhead is proportional for both algorithms. The Fibonacci underperformance therefore stems from pointer-chasing and cache-unfriendly consolidation, not from interpreter overhead.

B. Dial's Natural Pruning Mechanism

To understand why Dial's achieves up to $4.28\times$ speedup, we instrument its bucket operations in Experiment 8.

Unprocessed pushes. Let P denote total bucket pushes and S denote stale skips. In a lazy deletion heap, $P = V + S$ (conservation), so every pushed entry is either settled or skipped. In Dial's, however, $P > V + S$: pushes to future buckets beyond the maximum settled distance are *never popped at all*.

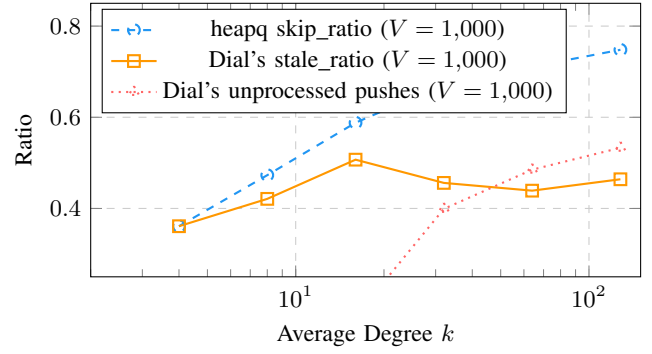


Fig. 6. heapq skip_ratio vs. Dial's stale_ratio and unprocessed-push fraction as functions of k (Experiment 8, $V = 1,000$, log x -axis). As k grows, heapq's skip_ratio climbs monotonically while Dial's stale_ratio plateaus; the gap is absorbed by unprocessed pushes.

We call these *unprocessed pushes* and denote their fraction $\rho_{\text{unp}} = (P - V - S)/P$.

Figure 6 tells the story. For $k = 64$, heapq skip_ratio is 0.711 and Dial's stale_ratio is only 0.439—a difference of 0.272. That difference is accounted for by unprocessed pushes ($\rho_{\text{unp}} = 0.485$): **nearly 49% of all Dial's pushes are never processed**, because the algorithm terminates once all V nodes are settled, leaving future-bucket entries abandoned. By contrast, the binary heap must pop every pushed entry.

The mechanism has an elegant explanation: in Dial's, settling node u at distance d pushes its neighbors into buckets at $d + w$, some of which may be far ahead of the current bucket pointer. When the nearer of u 's neighbors are settled later via better paths, those far-future entries become both stale *and* unreachable before the pointer reaches their bucket. Dense graphs (k large) produce more such “orphaned” entries. This is a qualitative advantage that binary heaps cannot replicate.

C. Integer vs. Float Weights

Experiment 9 repeats the Python and C++ comparison with float weights $w \sim \mathcal{U}[0, 1]$ (excluding Dial's, which requires integer weights). The relative ordering is unchanged: C++ Fibonacci is $1.6\text{--}2.7\times$ slower than heapq at $V = 10,000$, and the Python/C++ ratio is $15\times$ for both algorithms. We conclude that the performance relationship between heapq and Fibonacci heap is **independent of weight type**, generalizing our findings beyond integer graphs.

D. Data-Structure Selection Framework

Effect of maximum edge weight W (Experiment B). All prior comparisons used $W=100$. Experiment B varies $W \in \{100, 500, 1,000, 5,000, 10,000\}$ with $V=10,000$ fixed to characterize the interaction between W and Dial's speedup. For sparse graphs ($k=8$), Dial's advantage erodes sharply: speedup is $2.85\times$ at $W=1,000$ but falls to $0.80\times$ at $W=5,000$ and $0.66\times$ at $W=10,000$, with a breakeven threshold $W^* \approx 4,600$ (linear interpolation). For denser graphs ($k \geq 32$), Dial's remains faster than heapq even at $W=10,000$ ($\geq 1.07\times$ speedup), so $W^* > 10,000$ in this regime. The mechanism is the circular-array traversal cost: with $W+1$ buckets,

TABLE V

DATA-STRUCTURE SELECTION FRAMEWORK. $k_{\text{low}} \approx 10$ SEPARATES REGIMES WHERE $\text{SKIP_RATIO} < 0.5$. W^* IS THE MAXIMUM EDGE WEIGHT BELOW WHICH DIAL’S BEATS HEAPQ (EXPERIMENT B; $V=10,000$).

Condition	Recommended Rationale	
Integer weights, any k , $W \leq W^*$ [†]	Dial’s bucket queue	Up to $4\times$ speedup (sparse) via natural pruning; $O(V+E+W)$.
Integer weights, $k=8$, $W \gtrsim 5,000$	heapq	Dial’s speedup < 1 at $W=5,000$ ($0.80\times$) and $0.66\times$ at $W=10,000$; circular-array traversal dominates.
Float weights, $k \leq 10$ (sparse)	heapq (lazy deletion)	$\text{skip_ratio} < 0.5$; overhead acceptable; simplest implementation.
Float weights, $k > 10$ (dense)	heapq (lazy deletion)	Fibonacci does not outperform heapq even in C++; no benefit to switch.
Fibonacci heap	Avoid in practice	$1.5\text{--}2.3\times$ slower than heapq in C++ due to pointer-chasing; theoretical $O(V \log V + E)$ advantage is not realized.

[†] W^* depends on k : for $k=8$, $W^* \approx 4,600$; for $k \geq 32$, Dial’s remains faster even at $W=10,000$ ($\geq 1.07\times$ speedup), so $W^* > 10,000$ in this regime. See Section VI-D.

TABLE VI
FORMULA VALIDATION ON SNAP REAL-WORLD GRAPHS
(EXPERIMENT 10).

Dataset	V	E	k	Measured skip_ratio	Predicted (Eq. 1)
ego-Facebook	4,039	88,234	43.7	0.655	0.671
ca-GrQc	5,241	14,484	5.5	0.335	0.435
roadNet-CA	1,965,206	2,766,607	2.8	0.116	0.326
				Relative error	
ego-Facebook				2.5%	
ca-GrQc				23.1%	
roadNet-CA				64.4%	

the number of bucket-scan iterations grows proportionally to W , eventually dominating the $O(E \log E)$ savings.

Table V consolidates our findings into actionable guidance. The decision depends on two observable graph properties: weight type and average degree k .

VII. REAL-WORLD VALIDATION

We test formula (1) on three SNAP [12] datasets. Integer weights $w \sim \mathcal{U}_{\mathbb{Z}}[1, 100]$ are assigned randomly (seed 42); skip_ratio is averaged over 10 random source vertices. Results appear in Table VI and Figure 7.

ego-Facebook (2.5% error). This dense social network ($k = 43.7$) closely resembles an Erdős–Rényi graph: high average

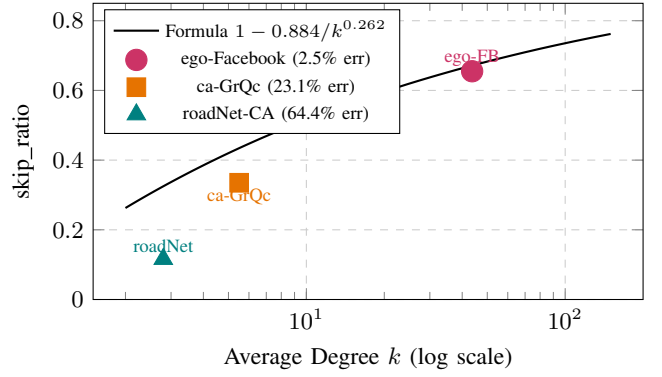


Fig. 7. skip_ratio formula validated on SNAP datasets (Experiment 10). Black curve: formula (1) fitted on random graphs. Colored markers: measured skip_ratio on real graphs.

degree, low clustering relative to k , and short cycles of length ≥ 4 . The locally tree-like assumption holds approximately, and the formula achieves 2.5% relative error—well within experimental noise.

ca-GrQc (23.1% error). The Arxiv collaboration network has $k = 5.5$, placing it at the left tail of the power-law curve. The error has two separable components. *Small- k component*: even on an ER random graph at $k \approx 5.5$, the formula carries $\Delta \approx 2\text{--}4\%$ inaccuracy (by interpolation from the $k = 4$ under-prediction of $\Delta \approx 0.03$); this accounts for roughly 2–4 percentage points of the 23% gap. *Structural component*: collaboration networks exhibit high clustering (many short cycles), violating the tree-like assumption on which the formula rests; the residual $\approx 19\text{--}21\%$ error is attributable to this structural mismatch. The two effects are therefore roughly 1:7 in magnitude.

roadNet-CA (64.4% error). This is not a limitation of the formula but is *consistent with the scope of its derivation*. Road networks are planar graphs (bounded genus, no long-range edges); their average degree $k = 2.8$ is far below the range for which the locally tree-like assumption holds, and their structure introduces long-range correlations in settling order. Measured skip_ratio is 0.116 vs. predicted 0.326—road networks exhibit markedly *less* heap waste than random graphs of the same degree because most paths are monotone along road corridors. Applying formula (1) to planar graphs would be a misuse; we treat the deviation as confirmation that the formula’s applicability boundary is correctly identified by the locally tree-like condition.

Scale-free (Barabási–Albert) graphs (Experiment C). Real social and technological networks often follow a power-law degree distribution rather than the Poisson distribution of Erdős–Rényi graphs. We test formula (1) on Barabási–Albert (BA) graphs [10] with attachment parameter $m \in \{4, 8, 16, 32\}$ (expected $k_{\text{avg}} = 2m \in \{8, 16, 32, 64\}$), $V \in \{1,000, 10,000\}$, random integer weights, and 10 repetitions per configuration.

The formula generalizes well to scale-free structure: at $k_{\text{avg}} \geq 16$, relative error is 2–4% for both BA and ER

graphs—essentially indistinguishable. At $k_{\text{avg}} \approx 8$ (sparse), BA graphs show a modestly lower `skip_ratio` (≈ 0.450) than ER graphs with matching k_{avg} (≈ 0.487), yielding 7–8% relative error instead of $< 6\%$. The gap is attributable to degree heterogeneity in BA graphs: hub nodes (high-degree) are settled early and generate fewer subsequent relaxations per unit of k_{avg} , slightly reducing the stale-pop fraction. Nonetheless, the formula provides a useful first approximation for scale-free graphs, extending its applicability beyond Erdős–Rényi random graphs.

VIII. DISCUSSION AND LIMITATIONS

Scope of validity. Formula (1) is calibrated on Erdős–Rényi-like random graphs with uniform edge weights. It generalizes to dense social networks (ego-Facebook, 2.5% error) but not to planar or hierarchical graphs. For structured inputs, the `skip_ratio` can be substantially lower than the formula predicts, as roadNet-CA demonstrates. A practitioner should verify that their target graph exhibits high average degree and low local clustering before applying the formula.

Fibonacci heap: theoretical vs. practical optimality. The Fibonacci heap achieves the theoretically optimal $O(V \log V + E)$ bound, yet our C++ measurements show it is $1.5\text{--}2.3\times$ slower than a binary heap for $V \leq 10,000$ across all tested densities. This gap is language-independent—we replicate it in both Python and C++ with the same ratio—and stems from pointer-chasing overheads during consolidation and cascading cuts. For graphs where $E \gg V \log V$ (i.e., extremely dense), the Fibonacci heap’s asymptotic advantage might eventually manifest, but such densities exceed the range studied here. We recommend against Fibonacci heaps unless theoretical guarantees are mandated by a formal proof system.

Structures not evaluated. This study compares three structures: binary-heap `heapq`, Fibonacci heap, and Dial’s bucket queue. Several practically important alternatives are not evaluated. *D-ary heaps* [13] reduce tree height to increase cache efficiency; their `skip_ratio` profile under lazy deletion mirrors binary `heapq` and falls within the same power-law characterization, but the constant factors differ and merit direct measurement. *Indexed binary heaps* (with a position array enabling $O(1)$ decrease-key lookup) eliminate stale entries entirely, avoiding `skip_ratio` overhead at the cost of random-access memory; they may outperform `heapq` on sparse float-weight graphs. *Pairing heaps* offer a simple decrease-key with favorable amortized cost; their lazy-deletion variant has not been characterized. Extending the selection framework to these structures is a natural next step.

Statistical note. Main experiments (Sections IV–VI) use 5–10 repetitions per (V, k) configuration and report means. Formal confidence intervals are not provided; however, the observed inter-run variation is ≤ 0.013 in `skip_ratio` across all (V, k) pairs (Table I), indicating high stability. Weight-distribution experiments (Section IV-D) use 30 repetitions ($V = 3,000$) or 20 repetitions ($V = 10,000$).

Future work. Three open questions arise naturally from this study:

- 1) *Theoretical derivation of the two-regime empirical model.* Section IV-D establishes empirically that the local slope $\alpha(k)$ converges toward $1/H_k$ and that the `push_count`/ V ratio approaches H_k from below as $k \rightarrow \infty$. A formal proof connecting the record-minimum process on Poisson–Galton–Watson trees to both the power-law regime and the harmonic limit—and a derivation of the crossover scale $k \approx 100$ —remains open.
- 2) *Exact characterization of $\alpha(F, W)$.* The dependence $\alpha \approx f(\log(W_{\text{max}}/W_{\text{min}}))$ is empirically clear but analytically unproven. An exact formula for the functional $\alpha(F, W)$ and its limiting value under continuous distributions would close this gap.
- 3) *Structured-graph models.* The 64% error on roadNet-CA motivates a separate characterization of `skip_ratio` for planar and hierarchical graphs, which have known algebraic structure amenable to analytic approaches.

IX. CONCLUSION

We have presented the first systematic empirical characterization of the lazy deletion overhead in Dijkstra’s algorithm as a function of graph structure. Our key findings are:

- 1) **Power-law `skip_ratio`, V-independent.** `skip_ratio`(k) $\approx 1 - 0.884/k^{0.262}$ ($R^2 = 0.984$) holds across graph sizes from 500 to 10^6 vertices, confirming that `skip_ratio` is primarily determined by average degree k under fixed weight distribution and ER-like graph topology. The formula generalizes to Barabási–Albert scale-free graphs with $< 8\%$ relative error (Experiment C).
- 2) **Two-regime empirical model and weight-distribution dependence.** The exponent α is not universal: it is strongly associated with $\log(W_{\text{max}}/W_{\text{min}})$ and ranges from 0 (constant weights) to ≈ 0.32 (continuous uniform, $k \leq 64$). At larger k , the local slope converges toward $1/H_k$, revealing that the power-law regime transitions to a harmonic regime $1 - A/H_k$ as $k \rightarrow \infty$ —consistent with the classical i.i.d. record-minimum bound and never violating it.
- 3) **Threshold fixes are broken.** Push-blocking heuristics cause propagation gaps corrupting up to 74% of shortest distances at any setting that yields measurable speedup.
- 4) **Fibonacci heap is not practical.** Despite $O(V \log V + E)$, Fibonacci heap is $1.5\text{--}2.3\times$ slower than `heapq` in optimized C++, language- and density-independent.
- 5) **Dial’s natural pruning delivers $4\times$ peak speedup.** Up to 49% of Dial’s bucket pushes are never processed—a structural property with no binary-heap analog—enabling **peak** $4.3\times$ speedup for sparse integer-weight graphs and a geometric mean of $2.9\times$ in C++ ($1.5\times$ in Python).

The practical takeaway is: *skip_ratio is a tight function of average degree k and weight distribution; matching graph structure to the right priority queue via our selection framework achieves up to $4\times$ peak speedup.* Specifically: use Dial’s for sparse integer weights, `heapq` for float weights, and avoid Fibonacci heaps in practice.

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